

Advanced Computer Networks (CS3.402)

Mid Semester Examination (Monsoon 2026)

International Institute of Information Technology, Hyderabad

Time: 1 hour and 30 minutes

Total Marks: 40

Instructions: Answer ALL questions.

This is a closed books and notes examination.

Write your answers sequentially as given in the question paper,
and also all the parts of a question at the same place.

No query is allowed in the examination hall.

Use of Regular Calculator is allowed.

22 September 2026

1. (a) Discuss, in brief, the evolution timeline for computer networks starting from 1950s (Centralized Mainframes).
(b) Explain, in brief, bluetooth master-slave architecture. Give some applications of Personal Area Networks (PANs).
(c) Discuss the structure and working of a fiber optic cable.

[3 + 4 + 3 = 10]

2. (a) Ten 9600-bps lines are to be multiplexed using TDM. Ignoring overhead bits in the TDM frame, what is the total capacity required for synchronous TDM? Assuming that we wish to limit average TDM link utilization to 0.9, and assuming that each TDM link is busy 60% of the time, what is the capacity required for statistical TDM?

(b) Explain the Bit Stuffing algorithms used at the transmitter and receiver sides for frame synchronization.

- (c) No mention was made of reject (REJ) frames in the stop-and-wait ARQ discussion. Why is it not necessary to have REJ0 and REJ1 for stop-and-wait ARQ?

[4 + 4 + 2 = 10]

3. (a) Determine the length of a contention slot in CSMA/CD for a 40 km multimode fiber-optic cable, where the signal propagation speed is 75% of the signal propagation speed in vacuum. Note that the speed of light in vacuum is $c = 3 \times 10^8$ m/s.

(b) For error correction of a frame used in the datalink layer, given the following parity-check matrix, H :

$$H = \begin{pmatrix} 1 & 0 & 1 & 0 & 1 & 0 & 1 \\ 0 & 1 & 1 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 & 1 & 1 & 1 \end{pmatrix}.$$

- ~~(i)~~ Explain why its null-space $N(H)$ is a single-error correcting code.
- ~~(ii)~~ What is the size of the (single-error correcting) code produced by H - explain.
- ~~(iii)~~ Decode the received tuple $\langle 0\ 1\ 1\ 0\ 0\ 1\ 0 \rangle$ assuming that error, if any, is a single-error.
- ~~(c)~~ Explain the pure ALOHA protocol in a flowchart (algorithm).

[3 + 4 + 3 = 10]

4. ~~(a)~~ Prove that, in Code-Division Multiple Access (CDMA), a receiving station can get the data sent by a specific sender if it multiplies the entire data on the channel by the sender's chip code and then divides it by the number of stations.
- ~~(b)~~ Explain, in brief, the working of Virtual Circuit Packet Switching. How is it different from the Datagram Packet Switching?
- ~~(c)~~ Suppose we use the Random Routing in a packet switching network. Let a node have four outgoing links with the link capacity of $R_1 = 10$ Mbps, $R_2 = 15$ Mbps, $R_3 = 20$ Mbps, and $R_4 = 5$ Mbps. Using this routing protocol, which link will be selected by the node?

[3 + 4 + 3 = 10]

***** End of Question Paper *****